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A network theoretic method for the basic reproductive number for infectious diseases

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Abstract

1. When an outbreak of an infectious disease occurs, whether it is COVID-19 in humans, pine wilt in trees or canine distemper in dogs, quick and decisive actions need to be taken to contain it. One tool that can help both the scientific and applied management communities understand the infection risk during an outbreak is the basic reproductive number, $\mathscr{R}_0$.
2. In this paper, we use a network method to calculate and analyse the basic reproductive number, specifically flow network theory. We convert traditional compartmental models to flow networks and then apply the fundamental Max-Flow Min-Cut Theorem to calculate the basic reproductive number.
3. We show that this method is equivalent to the traditional next generation matrix method for the calculation of $\mathscr{R}_0$, and thus a valid alternative. Then we provide step-by-step instructions and illustrate how to apply this method to epidemic models.
4. The current methods available for calculating $\mathscr{R}_0$ are complicated, requiring mathematical training. This can act as a barrier to understanding and cause delays in

community. It also allows for novel insights and can be applied to models/situations where traditional methods fail.

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One of the most influential developments in epidemiology, public health and wildlife management is the characterization of the basic reproductive number, often referred to as $\mathscr{R}_0$. While originating in the fields of demography and epidemiology (Heesterbeek, 2002), the basic reproductive number has become an important aspect of mathematical biology and ecology and is now mathematically defined and quantifiable (Diekmann et al., 1990; Dietz, 1975; van den Driessche & Watmough, 2002). Epidemiologically, $\mathscr{R}_0$ is defined as the expected number of secondary infections produced by one infectious index case in a completely susceptible population (Diekmann et al., 1990), while mathematically $\mathscr{R}_0$ is defined as the threshold parameter for invasion of an infectious pathogen into a completely susceptible population (Brauer & van den Driessche, 2008). Calculating $\mathscr{R}_0$ offers important information in responding to an outbreak as its value can help determine if a disease can invade a population. If $\mathscr{R}_0 > 1$, the infectious individuals will replace themselves and increase with each generation, but if $\mathscr{R}_0 < 1$, infectious individuals will fail to replace themselves and thus the disease will die out. In addition to determining whether a disease will successfully take hold in a population, the value of $\mathscr{R}_0$ gives an idea of how quickly disease will initially spread. A larger value for $\mathscr{R}_0$ implies the disease will spread quicker, like measles, which some studies show can have a basic reproductive number greater than 50 (Guerra et al., 2017), while a smaller $\mathscr{R}_0$ points towards a slower spread, as seen in canine rabies which has reported $\mathscr{R}_0$ values of just above 1 (Ferguson et al., 2015). Thus, mathematical analysis of $\mathscr{R}_0$ is an important step in both describing and predicting a disease outbreak and in formulating a response.

Analysing $\mathscr{R}_0$ can also provide critical insight on how to decrease its value and therefore decrease the spread of infection. This type of analysis is very commonly seen in the mathematical biology, ecology and epidemiology literature (Anderson et al., 1981; Dantzler et al., 2016; Khan et al., 2017; Liu et al., 2020; Lloyd et al., 2020; Mitchell & Kribs, 2017; Ruan, 2017b; Smith et al., 2007; Valega-Mackenzie & Ríos-Soto, 2018; Wiraningsih et al., 2015). Depending on the model type there are different ways to calculate $\mathscr{R}_0$. The most basic way is through visual inspection. This method will usually only work for simple susceptible-infected-recovered (SIR)-type models by determining what makes the infectious equation, or $\dot{I}$, positive near the disease free equilibrium (DFE) and therefore the infectious class increasing. This method leads to the definition of the basic reproductive number as: (duration of infectious period) × (the rate of secondary infections). While this is a quick and easy way to determine $\mathscr{R}_0$, it

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et al. (1990) and built upon by van den Driessche & Watmough (2002), allows for $\mathscr{R}_0$ to be

Chavez, 2020; Mitchell & Kribs, 2017). While these two methods are the most common to find the basic reproductive number, there are additional methods discussed in Li et al. (2011).

One approach to understanding and calculating $\mathscr{R}_0$ discussed in Li et al. (2011) is through graph theory. Graph, or network, theory is a branch of mathematics that can offer a different perspective on disease dynamics and modelling (Roberts, 1978). A graph, also referred to as a network, is comprised of two sets, vertices and edges. Vertices, also called nodes, are connected to each other by edges. This map of connections of vertices by edges is called a graph or network. Networks are often represented visually by a digraph (Figure 1).

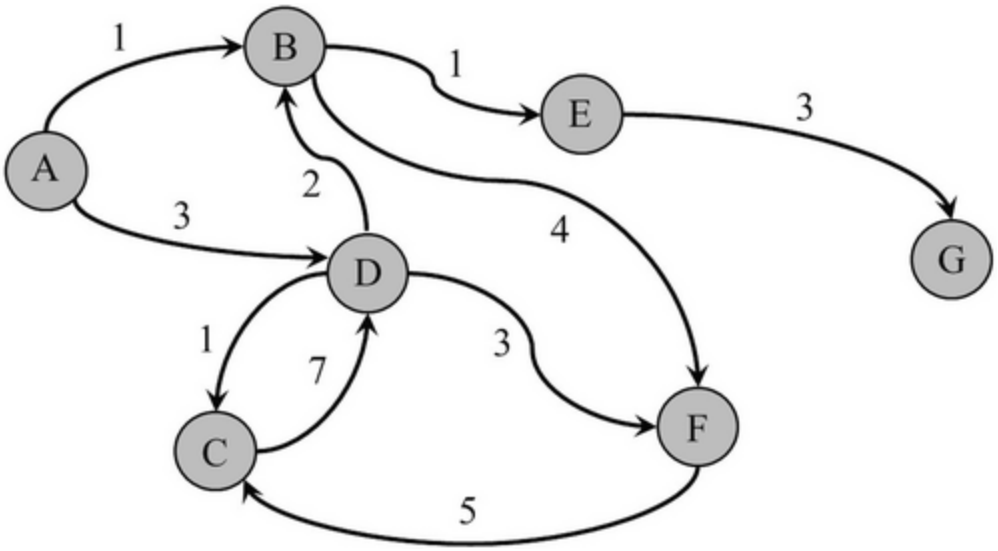


FIGURE 1

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Example of a weighted digraph where the circles with the letters inside are the nodes, the arrows are the edges, and the numbers above the edges are the edge weights

Network models offer insight to disease transmission dynamics that may be overlooked when using other model types. They can be especially helpful in explaining mathematical models and results to a nonmathematical audience, as networks can be more intuitive than other model types (e.g. ordinary or partial differential equations). Networks have been used to study

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to learn about general epidemic dynamics (Keeling & Eames, 2005). Thus, graph theoretic approaches have already been explored as a way to learn about the basic reproductive

calculate $\mathscr{R}_0$. Network theory is also used to analyse $\mathscr{R}_0$, such as in (Koch et al., 2013), which studies the effect that edge removal has on the size of $\mathscr{R}_0$.

Thus far, we have discussed two of the main ways to model infectious diseases. First, is using compartmental ordinary differential equation (ODE) models. This approach tracks how population segments move through health states. This is a community-level approach to modelling. To model infectious disease spread on the individual level, network models can be used. Networks model how individuals infect each other. We are proposing a third method to look at infectious disease spread: modelling how infection itself flows through a network of population segments. Using this third method we develop a network theoretic approach to calculating and analysing $\mathscr{R}_0$. To do this we use a specific type of network model—a flow network. A flow network model (sometimes called a transportation network) is a directed graph in which the nodes and edges have capacities to handle 'flow' through them. Flow networks also have a source node and a sink node. A source node is the point of origin for all flow into the network and the sink node is where the flow terminates. No flow within the network may enter the source or exit the sink. For an epidemiological model, we will consider the disease to be the flow and the infectious class to be the source. (Note the slight difference in framing from standard ODE SIR models, in which individuals are the objects moving through the system (Brauer et al., 2008).) To use a flow network to compute the basic reproductive number for a disease, we therefore must first convert a traditional ODE compartmental model into a flow network. To do this, we consider each class as a node in the network and study the 'flow' of disease through the nodes. In this way, we develop an alternative approach to calculating the basic reproductive number and show this new method is equivalent to the NGM method for finding $\mathscr{R}_0$. We also present an algorithm for converting compartmental ODE models to flow networks and give instructive examples of how to use this method to calculate the basic reproductive number.

2 MATERIALS AND METHODS

2.1 Max-flow min-cut method

In order to calculate the basic reproductive number with a flow network, we need to determine the maximum amount of flow, or disease, that can be sent from the source to the sink. In an epidemiological context, this will be the maximum amount of disease generated from one i
t
words, the amount of flow that ends up in the sink represents the number of infections there will be in the next generation. To calculate this we need to solve the max flow problem, which is

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the Max-Flow Min-Cut Theorem, is equal to the max flow.

Theorem 1. For any flow network the value of the maximum flow from the source to the sink is equal to the weight of the minimum cut over all cuts that separate the source and sink. (Ford & Fulkerson, 1956).

Thus, once we determine the max flow, we can analyse it in a similar way to the traditional methods for finding the basic reproductive number. If the flow is positive, then the disease will continue to spread and, if it is zero or negative, then the disease dies out. Hence we will call this new method for calculating $\mathscr{R}_0$ the Max-Flow Min-Cut (MFMC) method. Note that the main difference between modelling an outbreak as a network versus a compartmental ODE model is the network model tracks the movement of infection in a population, while the ODE model tracks the movement of population segments through health stages.

2.2 Notation

Before we can show that the MFMC method is equivalent to the NGM method we must first define notation. We start with some general network notation. Let $N = (V, E)$ be a directed network where V is the set of all the vertices and $E = \{(u, v) \mid \forall u, v \in V\}$ is the set of all the directed edges (also called 'arcs'). Furthermore, let $w(u, v)$ be the weight of the (u, v) edge. Specifically for flow networks, we denote the source as $s \in V$ and the sink as $t \in V$. Now we can develop notation for our application. Note that the edges originating from s represent new infections. Denote this set of edges as Φ. Then denote the rest of the edges as Ψ. Thus, for a network, N, formulated from a compartmental ODE model $E = \Phi + \Psi$. Since we have notation for the edges, let us define notation for the edge weights. Let $w(\Phi)$ represents the total weight of all the edges in Φ and $w(\Phi_{u,v})$ represents the weight of the specific edge, $(u, v) \in \Phi$. We will use the same notation for the edges in Ψ.

For the next generation matrix we employ the notation from Brauer et al. (2008) and van den Driessche and Watmough (2002). Let x represents the disease compartments and let y represents the infection-free compartments. For this calculation, we are only interested in the infected compartments. Thus, we can write the infected compartments of the model as follows,

$$x_i' = \mathscr{F}_i(x, y) - \mathscr{V}_i(x, y), \quad i = 1, ..., n,$$

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$= \mathscr{V}^- - \mathscr{V}^+$ is the rate of disease removal ($\mathscr{V}^-$) and progression ($\mathscr{V}^+$) that decrease the ith

are the expected times an individual initially introduced into the infection compartment j spends in infection compartment i, the (i, j) entries of F are the rate secondary infections are produced in compartment i by an index case in compartment j, and so the (i, j) entries of FV^{-1} are the expected number of secondary infections in compartment i produced by individuals initially in compartment j (Table 1).

TABLE 1. Summary of the notation

Symbol	Description
Max-Flow Min-Cut Method	
N	Directed network
V	Set of all the vertices
E	Set of all edges
$w(x)$	Weight of x
s	Source node
t	Sink node
Φ	Set of edges representing new infections
Ψ	Set of the rest of the edges
Next Generation Matrix Method	
x	Disease compartments
y	Infection-free compartments
$\mathscr{F}_i$	Rate of secondary infections that increase the $i\text{th}$ infected compartment
$\mathscr{V}_i$	Rate of disease removal and progression that decrease the $i\text{th}$ compartment
FV^{-1}	Next generation matrix

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MFMC method and the NGM method provide equivalent calculations for the value of $\mathscr{R}_0$. First, let us consider how the next generation matrix (FV^{-1}) and the network (N) model secondary infections and their spread. In the NGM method, $\mathscr{F}$ is the rate at which secondary infections increase the infection compartments. In the MFMC method, Φ is the set of edges that represent the rates at which secondary infections spread to the rest of the nodes. Thus, $\mathscr{F}$ and Φ are qualitatively equivalent. The same is true for $\mathscr{V}$ and Ψ as they both represent the rest of the movement in the infection compartments.

In addition to both methods representing the same qualitative ideas, they are also quantitatively equivalent with some regrouping. Note,

$$w(\Phi) = \mathscr{F} + \mathscr{V}_I^-$$
$$w(\Psi) = \mathscr{V}_{i^*}^+ + \sum_{j \neq I} \mathscr{V}^-,$$

where $\mathscr{V}_I^-$ represents removal terms from the I class, $\sum_{j\neq I}\mathscr{V}^-$ represents the rest of the removal terms and $\mathscr{V}_{i^*}^+$ represents all of the progression terms except those that allow for t to feed directly into s. Note, the exclusion of these terms does not cause a difference in $\mathscr{R}_0$, instead they are essentially functioning as lag terms (which do not affect the contribution of new infections towards the next generation). These lag terms also disappear while computing FV^{-1}, see Section 5. Thus, $\mathscr{F}, \mathscr{V}$ and $\Phi, \Psi, w(\Phi), w(\Psi)$ are equivalent in both mathematical expression and interpretation.

Next, the NGM method computes $\frac{\partial \mathscr{F}}{\partial x}\big|_{(0,y_0)} = F$ and $\frac{\partial \mathscr{V}}{\partial x}\big|_{(0,y_0)} = V$ to consider the behaviour of the system at the beginning of an outbreak. The MFMC method also considers the system behaviours near the DFE, but does so by evaluating $N|_{(S^*\approx P, I^*>0)}$, where P is the total population and I^* is small. (Note this is the same method used when calculating $\mathscr{R}_0$ by inspection.) While these methods for analysing behaviours near the DFE are different, they are equivalent (for further information see Van den Driessche (2017) for an example of when both methods are used in finding the basic reproductive number and Strogatz (2018) for a more mathematically technical explanation of the equivalence).

Since we have seen that FV^{-1} and N have equivalent methods for modelling the amount of s
t
method $\mathscr{R}_0 = \rho(FV^{-1})$, where ρ is the spectral radius. The MFMC method calculates $\mathscr{R}_0$ by finding the minimum cut, which by the Max-Flow Min-Cut Theorem, is the maximum flow.

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equivalently, calculating $\mathscr{R}_0$ by finding dominant influence of the system by the spectral radius is equivalent to finding dominant influence in a network by maximum flow.

From these arguments, we see that the Next Generation Matrix method and the Max-Flow Min-Cut method are equivalent in both their understanding and calculation of the secondary infections at the beginning of an outbreak.

3 RESULTS

3.1 Algorithm and examples

To get a better understanding of both techniques and their equivalence, we present each algorithm for execution and use both to work through some examples. Consider a closed system, compartmental epidemic model. Assume a disease well described by a basic SIR model (Brauer et al., 2008),

$$\begin{aligned} \dot{S} &= -\beta SI \\ \dot{I} &= \beta SI - \gamma I \\ \dot{R} &= \gamma I, \end{aligned} \tag{1}$$

where β is the contact rate and probability of transmission of infection between the susceptible and infectious people and γ is the removal rate for the infectious class (either through recovery or death). Furthermore, let N be the total population (N will be the total population for all the following examples as well). See Table 2 for a summary of the symbols and parameters used in model 1. Due to the simplicity of the model we can compute $\mathscr{R}_0$ by inspection. First, we evaluate the model near the DFE ($S^* \approx N, I^* > 0$, but small) and then solve for $\dot{I} > 0$. Thus, the basic reproductive number is ${}^{\beta S^*}/_{\gamma}$. For instructive purposes, we now present the respective algorithms for finding $\mathscr{R}_0$ using both the NGM and MFMC methods, step by step.

TABLE 2. Symbol and parameter table for model 1

Symbol/parameter	Description

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R	Recovered population

N	Total population
β	Contact rate and probability of transmission of infection
γ	Removal rate

Next Generation Matrix Method:

a. Divide the classes into two compartments: disease, x and nondisease, y. We are only interested in the disease compartment.

$$x = I$$

$$y = S, R$$

b. Determine which terms from the equations for the classes in the disease compartment represent new infections in the disease compartment, $\mathscr{F}$, and which represent movement among, $\mathscr{V}^+$, or out of the disease compartment, $\mathscr{V}^-$.

$$\mathscr{F} = \beta SI$$

$$\mathscr{V} = \gamma I$$

c. Take the derivative of $\mathscr{F}$ and $\mathscr{V}$ with respect to the state variables of the disease compartment, x. We call the resulting matrices F and V.

$$\frac{\partial \mathscr{F}}{\partial x} = \beta S = F$$

$$\frac{\partial \mathscr{V}}{\partial x} = \gamma = V$$

d. Evaluate F and V at the DFE.

DFE: $(S^*,0,0)$

$$F|_{(S^*,0,0)} = \beta S^*$$

$$V|_{(S^*,0,0)} = \gamma$$

e. γ

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f. Determine $\rho(FV^{-1}) = \mathscr{R}_0$. This, along with its eigenvector represents the distribution of infected individuals that produces the greatest number of secondary infections per generation.

$$\mathscr{R}_0 = \frac{\beta S^*}{\gamma}$$

Max-Flow Min-Cut Method:

1. Divide the compartments into two groups: compartments through which infection can flow and compartments through which it cannot. We are only interested in the compartments that act as conduits for infection propagation. These compartments will become the nodes in our network.

   Infection flows through: I, S

   Infection does not flow through: R

2. Identify the infectious class(es). This is the compartment(s) that can transmit the infection. This compartment will be the source node, s.

   $s = I$

3. Identify the compartment(s) that connects, or feeds directly, into the infectious class(es). This compartment will be the sink node, t.

   $t = S$

4. Determine the edges and identify the edge weights. This step is very model dependent.

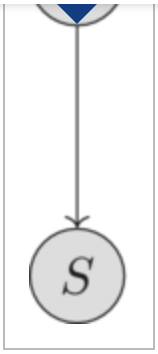
4.a The remaining compartments will be the intermediary nodes in the network. All of the nodes in the network are connected by edges. The set of edges that leave the source node and represent the new infections in the community make up Φ. While the rest of the edges that represent the movement of the infection are in the set Ψ.

$\Phi = \{(I, S)\}$

$\Psi = \emptyset$

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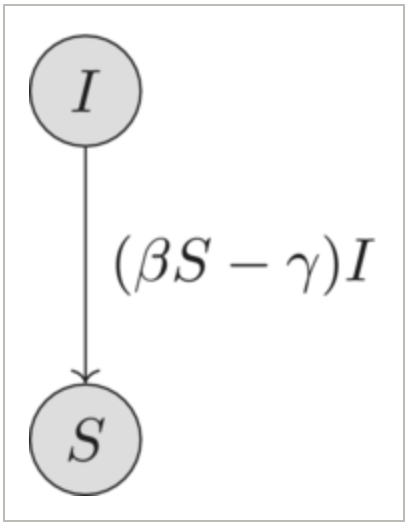
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4.b The weights of the edges, $w(\Phi), w(\Psi)$, come directly from the terms in the compartmental model. Depending on the structure of the model, some edges may have repeated weights due to conservation of flow. Since this is a population-level network, net flow of disease also needs to be considered.

$$w(\Phi) = (\beta S - \gamma)I$$

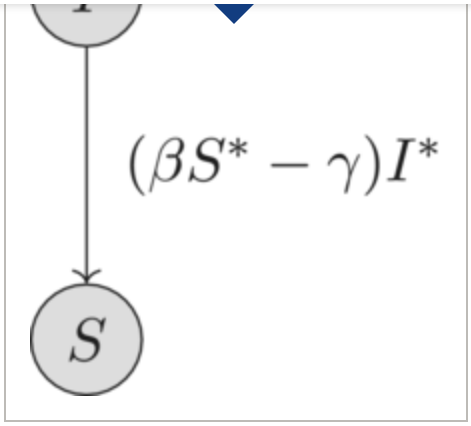
$$w(\Psi) = \emptyset$$



5. The compartmental ODE model is now formulated into a network model. Next we evaluate the edges near the DFE. In other words, we let the infectious class be small, but positive, and the susceptible class be approximately the entire population: ($S^* \approx N, I^* > 0$, but small).

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6. Find the cut sets, which are the sets of edges that, when cut, separate s from t. The cut sets represents the ways the disease can spread through a community to cause secondary infections and the cuts show how the spread can be stopped. The capacity of the cut represents the amount of disease/infection available for secondary infections.

   Cut set: $\{(I, S)\}$

7. Determine the cut set with the minimum edge weight. By the Max-Flow Min-Cut Theorem, we know that is also the max flow.

   Minimum cut weight = Maximum flow: $(\beta S^* - \gamma)I^*$

8. Solve the equation for the epidemic growth condition desired. For example, for an outbreak to occur (i.e. positive growth), the max flow must be positive, so we can solve the inequality for $\mathscr{R}_0 > 1$.

   Max flow $> 0 \Rightarrow (\beta S^* - \gamma)I^* > 0 \Rightarrow \mathscr{R}_0 = \frac{\beta S^*}{\gamma} > 1$

Now that we have defined the algorithms and worked through a concrete example implementing both methods, we can gain better intuition into the MFMC method by examining the step-wise equivalence between the methods. First, we note that both techniques start off similar by considering where there is infection in the model. A subtle difference is that while the NGM method looks at where the infected (not just infectious) individuals are in the compartmental model, the MFMC method looks at which compartments house infection and where it can spread from there. Thus, 1. and (a) are equivalent. Next, both techniques work to i

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We note that in the MFMC method, the edges that leave the source play the same role that $\mathscr{F}$ does in the NGM method, and the rest of the edges in the network represent the same type of

by linearizing about the DFE, ((c)–(d)) while the MFMC method does this by analysing what the system does when it is just off the DFE, (5.). This is similar to the more basic technique for solving for $\mathscr{R}_0$ by finding which parameter combinations make $I^{\cdot} > 0$ near the DFE. Finally, the interpretations of the entries FV^{-1} and its eigenvalues/vectors (including $\rho\left(FV^{-1}\right)$) in (e)–(f) and the interpretations of the cut sets, edge capacities and max flow seen in 6.–7. consider the same idea from different perspectives. The NGM method considers how infected and infectious individuals move through the classes and produce secondary infections, while the MFMC method considers how the infection moves through the nodes and how many infections there will be to contribute towards new infections in the next generation.

Keeping this step-wise equivalence in mind, let us consider the example. Since the NGM method is an established technique, we will focus on how the MFMC algorithm was applied. First, we note that the only class that is not explicitly included in the network is the R class, as disease cannot flow into it. However, the movement of infectious individuals into the R class does have an effect on the amount of infection in the network. Since infectious individuals are moved to the R class at rate γ, that is also the rate at which infection is removed from the network. Thus, since the network is modelling a community, not an individual, we need to take γ into consideration when determining the overall amount of infection that can flow into the community. Hence, the amount of infection that can flow from the infectious node to the rest of the network is the amount of effective contacts minus the amount that has been removed/recovered, which can be thought of as the net flow of new disease. So, the weight of the only edge in the network is $\beta S^* I^* - \gamma I^*$, which is equal to $\mathscr{F} - \mathscr{V}$. Figure 2 provides supplemental information and explanations on how to set up the network and assign the edge weights for model 1.

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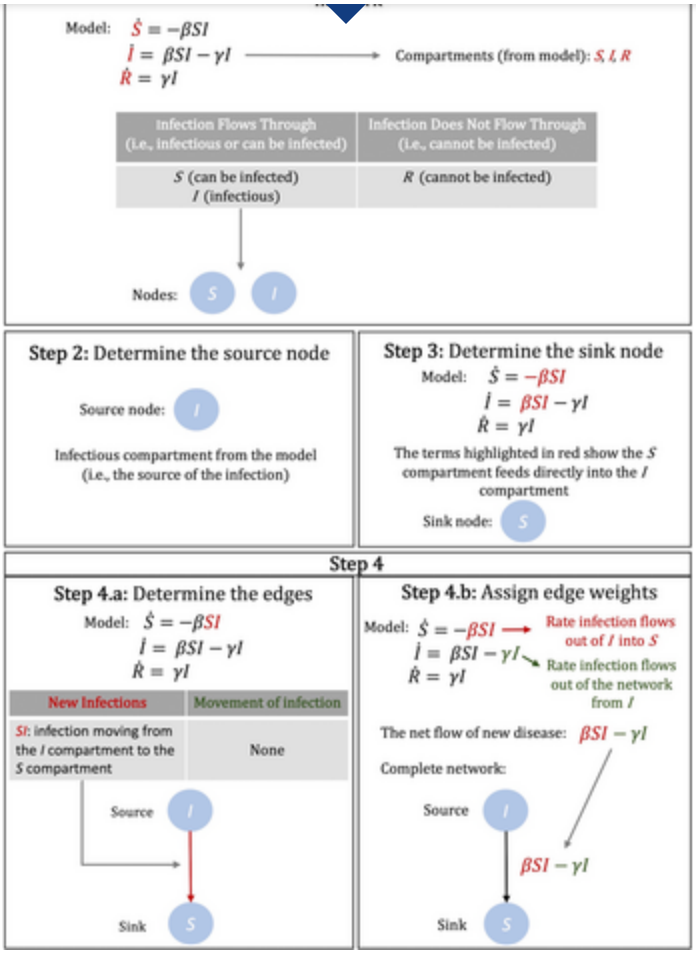


FIGURE 2

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Diagram for steps 1–4 (setting up the network) of the MFMC method for model 1

Next, let us consider a model with slightly more complexity by adding in an exposed class. The $SEIR$ model (Brauer et al., 2008) is,

$$\begin{aligned} \dot{S} &= -\beta SI \\ \dot{E} &= \beta SI - \kappa E \\ \dot{I} &= \kappa E - \gamma I \end{aligned} \tag{2}$$

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example, determining the basic reproductive number by inspection fails due to the inclusion of the exposed class. So, for more complicated models, a method like the NGM or MFMC is the most frequent way employed to analytically determine $\mathscr{R}_0$. The calculations below follow the same steps as defined above in each of the algorithms.

TABLE 3. Symbol and parameter table for model 2

Symbol/parameter	Description
S	Susceptible population
E	Exposed population
I	Infectious population
R	Recovered population
β	Contact rate and probability of transmission of infection
κ	Disease progression rate
γ	Removal rate

Next Generation Matrix Method:

a. $x = I, E$

$y = S, R$

b. $\mathscr{F} = \begin{pmatrix} \beta SI \\ 0 \end{pmatrix}$

$\mathscr{V} = \begin{pmatrix} \kappa E \\ \gamma I - \kappa E \end{pmatrix}$

c. $\frac{\partial \mathscr{F}}{\partial x} = \begin{pmatrix} 0 & \beta S \\ 0 & 0 \end{pmatrix} = F$

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d. DFE: $(S^*,0,0,0)$

$$V|_{(S^*,0,0,0)} = \begin{pmatrix} \kappa & 0 \\ -\kappa & \gamma \end{pmatrix}$$

e. $V^{-1} = \begin{pmatrix} \frac{1}{\kappa} & 0 \\ \frac{1}{\gamma} & \frac{1}{\gamma} \end{pmatrix}$

$$FV^{-1} = \begin{pmatrix} \frac{\beta S^*}{\gamma} & \frac{\beta S^*}{\gamma} \\ 0 & 0 \end{pmatrix}$$

f. $\rho(FV^{-1}) = \mathscr{R}_0 = \frac{\beta S^*}{\gamma}$

Max-Flow Min-Cut Method:

1. Infection flows through: I, S, E

Infection does not flow through: R

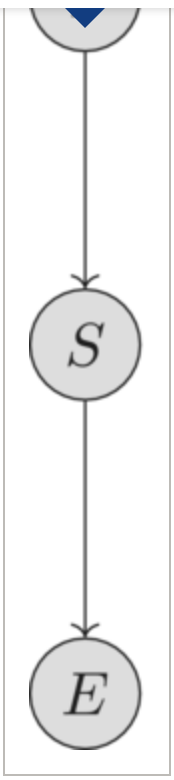
2. $s = I$
3. $t = E$

4.a $\Phi = \{(I, S)\}$

$\Psi = \{(S, E)\}$

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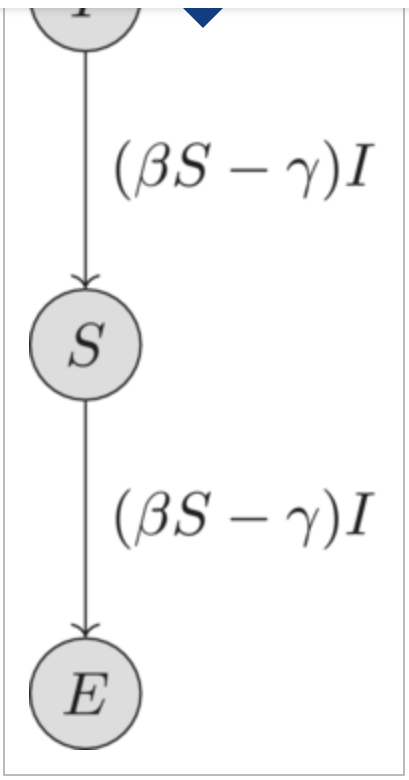


4.b $w(\Phi) = (\beta S - \gamma)I$

$w(\Psi) = (\beta S - \gamma)I$

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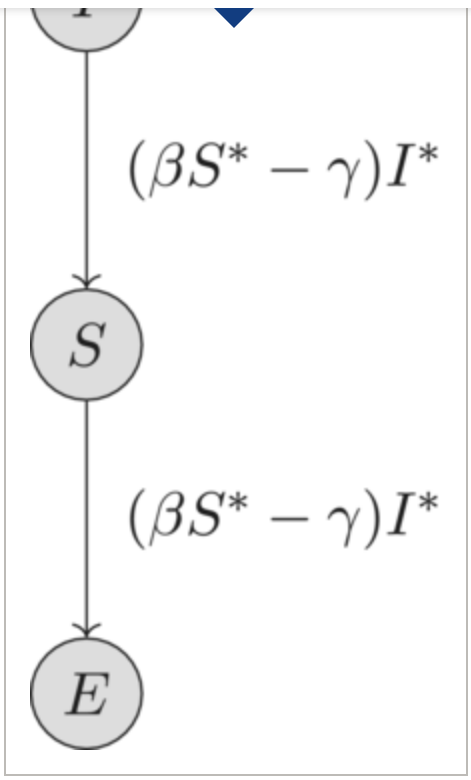
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5. Evaluate near DFE, $(S^*, I^*, 0)$:

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6. Cut set: $\{\{(I, S)\}, \{(S, E)\}\}$
7. Minimum cut weight = Maximum flow: $(\beta S^* - \gamma)I^*$
8. Max flow $> 0 \Rightarrow (\beta S^* - \gamma)I^* > 0 \Rightarrow \mathscr{R}_0 = \frac{\beta S^*}{\gamma} > 1$

The setup of the network model in this example is similar to the first and so we still must take into account the net flow of new disease (hence the γ term in the edge weights). The obvious difference is the extra node. Now the sink is the exposed class and the susceptible class is an intermediary node. Following the algorithm, the exposed class is the sink since it feeds directly into the infectious class (source) in the compartmental model. Since the network is set up such that the amount of infection that flows to the sink is the amount available for the source in the next generation and the sink and source cannot be connected, the κ term is ignored in the network. For the supplemental diagram on setting up the network for model 2, see Figure 3 in Appendix S1.

4 DISCUSSION

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cut sets that come from the algorithm. For the first example, since the model does not have much complexity, there is only one cut set, $\{(I, S)\}$. Hence the only way for disease to move

is only one cut set, its weight, $(\beta S^* - \gamma)I^*$, is the minimum and thus is the max flow, which represents the maximum amount of disease to flow through the network that would be available to produce secondary infections in the next generation of infectious individuals. Similarly for the NGM method, our 'matrix' is just a number, $\beta S^*/\gamma$, which represents the greatest (maximum) or expected number of secondary infections in the infectious compartment produced by individuals initially in the infectious compartment.

In the second example we have two cut sets: $\{(I, S)\}, \{(S, E)\}$. Even though there is still only one path from the source to the sink $(I \to S \to E)$, there are two places where the flow could be stopped. The first is to separate the infectious from the rest of the population, $\{(I, S)\}$. In real life this is done by isolating the sick, so they can no longer spread the disease to the susceptible population. The second cut set $\{(S, E)\}$, looks at quarantining anyone that has been exposed to the disease before they become infectious and thus preventing them from causing any new infections. In practice these cut sets recommend similar, yet critically distinct courses of action: quarantine and isolation. In theory, the second cut set represents a better option for disease control because if an individual is quarantined before they become infectious they will not be able to spread the disease, but if isolation only occurs after they are infectious they could have already spread the disease to others before isolation begins. Although better in theory, this option is much harder to implement in practice as contact tracing has many real world limitations.

Since both cuts occur on the only path that connects the source and the sink and no disease is lost along the path, they both have the weight of $(\beta S^* - \gamma)I^*$. Note these repeated edge weights are seen in models with simple and direct health stages due to the conservation of flow (as seen in this and the following examples in the next section and Appendix S1). Thus, this is the max flow and its interpretation is the same as in the first example. Using the NGM method, we note that the expected number of secondary infections in compartment E produced by individuals initially in compartment E and the expected number of secondary infections in compartment E produced by individuals initially in compartment I represent the same idea as weights of the cut sets and are thus both $\beta S^*/\gamma$. Then since all the other entries of FV^{-1} are 0, $\beta S^*/\gamma$ is the expected number of secondary infections in the next generation.

While the NGM method is a robust and widely used method for computing the basic reproductive number, the MFMC method is a useful alternative. By relying solely on a network

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understood by, a nonmathematical audience, this graphical approach is easier to explain and requires less foundational mathematical knowledge to understand. Thus, using a network-

own, instead of waiting for mathematicians, which can quicken response time for incipient outbreaks. Its intuitive nature allows for a better understanding of the concept of the basic reproductive number, so it can be used more effectively to track and prevent disease spread. Moving forward, the MFMC method and other network-based methods can be modified and formulated to help provide accessible insight on more complex disease dynamics. For example, the following section shows how to extend the MFMC method to include additional model characteristics.

In addition to providing a more accessible, alternative approach to understanding the basic reproductive number, the MFMC method also gives insights that the NGM method does not. As discussed above, the MFMC method offers information about the dynamics of the system and different methods of preventing an outbreak from progressing by analysing the cut sets. More intriguingly, however, it can also be applied to systems that have nonunique endemic equilibria and even provide nonequilibrium behaviour estimates by employing continuous time dynamics (Philpott, 1990) and continuous dynamics with delays (Ogier, 1988) in the flow network. By alleviating the constraints imposed by having to solve by substitution at an equilibrium point, these methods can allow much more nuanced characterization of transient behaviour. As the field of network theory itself continues to develop, with topics like machine learning and neural networks, it also has the potential to expand as a tool to work alongside traditional mathematical techniques to increase intuitive understanding, especially for nonmathematicians. By continuing to consider well known problems or techniques through the lens of network theory, we will be able to gain new insights and, hopefully, a better understand of the system we are studying.

5 EXTENSIONS OF THE MFMC METHOD

Beyond the direct method for the simple cases presented, natural extensions of the MFMC method can be used for more complicated models. We can expand the MFMC method to work for models with increased complexities. Let us consider the following $SEIR$ model with vital dynamics. Let $N = S + E + I + R$ and μ be the birth and death rate. Then we define the model as,

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$$\begin{aligned}\dot{E} &= \frac{\beta S I}{N} - (\mu+\kappa)E \\ \dot{I} &= \kappa E - (\gamma+\mu)I \\ \dot{R} &= \gamma I - \mu R\,.\end{aligned} \tag{3}$$

See Table 4 for a summary of the symbols and parameters used in model 3. The NGM method tells us that $\mathscr{R}_0 = \frac{\kappa}{\mu+\kappa}\frac{\beta}{\mu+\gamma}$ (Note, in the Data Availability Statement section a link to code is provided to compute $\mathscr{R}_0$ using the NGM method for models 1–3). This basic reproductive number has two components: $\frac{\kappa}{\mu+\kappa}$ and $\frac{\beta}{\mu+\gamma}$. The first factor is the probability of surviving the exposed class and going on to the infectious class and the second factor is the probability of transmitting infection through contact during the infectious period. The MFMC method easily computes the second factor by adjusting the edge weights to include the μ term from the I class, as it functions the same as the removal to the R class, γ, term. The MFMC method must be appended to recover the first factor. Note, since the MFMC method is based on a flow network it does not allow for flow (disease) to leave the sink or, due to the conservation of flow, escape the network. Thus, calculating how much diseases 'survives' the sink to be available to the source for the next generation must happen outside of the flow network. Luckily, determining this survival probability comes straight from the differential equation for the sink class. It has the form:

$$\frac{\text{rate of transition to source}}{\text{rate of transitioning out of sink}},$$

where transition out of the sink includes transition into the source. Thus, multiplying the result of the MFMC method by this fraction recovers $\mathscr{R}_0$. Note, this adjustment is consistent with the previous examples as without death terms the probability of surviving the sink is 1.

TABLE 4. Symbol and parameter table for model 3

Symbol/parameter	Description
S	Susceptible population
E	Exposed population
I	Infectious population

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R	Recovered population
N	Total population ($S + E + I + R$)
μ	birth and death rate
β	Contact rate and probability of transmission of infection
κ	Disease progression rate
γ	Removal rate

Next we will compute the basic reproductive number for model 3 using the MFMC method with the adjustment described above.

Max-Flow Min-Cut Method:

1. Infection flows through: I, S, E

Infection does not flow through: R

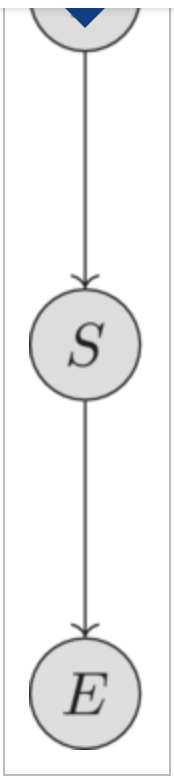
2. $s = I$
3. $t = E$

4.a $\Phi = \{(I, S)\}$

$\Psi = \{(S, E)\}$

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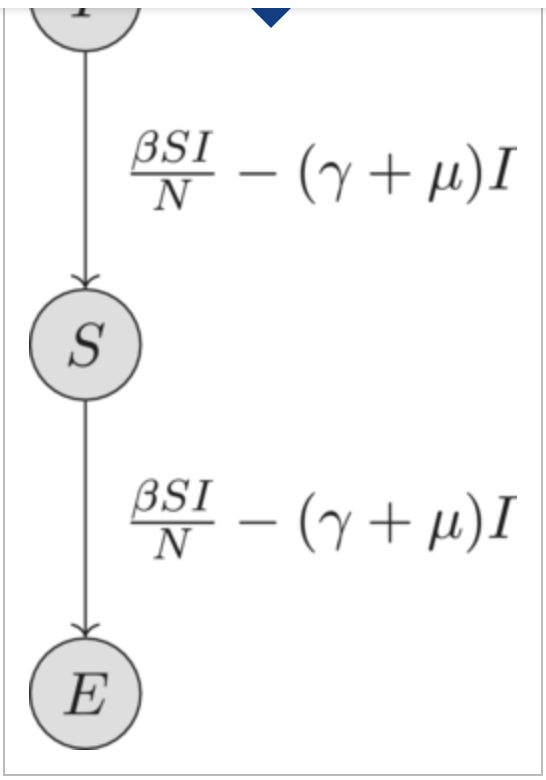


4.b $w(\Phi) = \frac{\beta SI}{N} - (\gamma + \mu)I$

$w(\Psi) = \frac{\beta SI}{N} - (\gamma + \mu)I$

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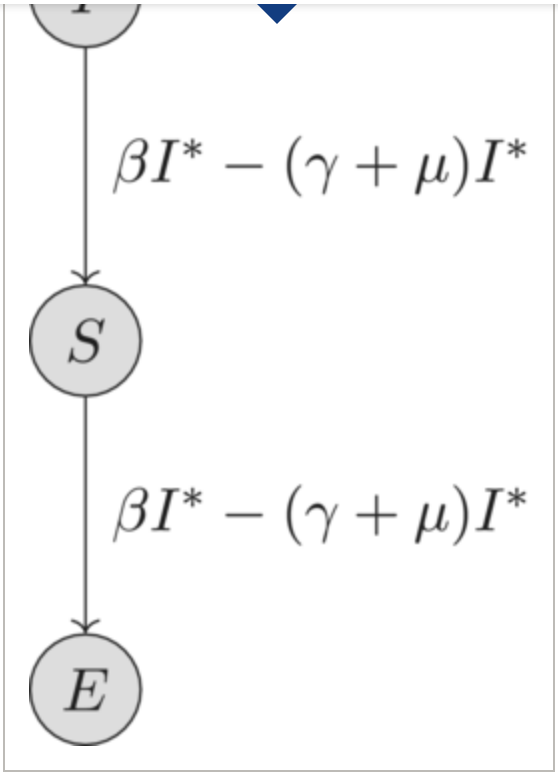
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5. Evaluate near DFE, $(S \approx N, I^* > 0, 0)$:

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6. Cut set: $\{\{(I, S)\}, \{(S, E)\}\}$
7. Minimum cut weight = Maximum flow: $\beta I^* - (\gamma + \mu) I^*$
8. Max flow $> 0 \Rightarrow \beta I^* - (\gamma + \mu) I^* > 0 \Rightarrow \frac{\beta}{\gamma + \mu} > 1$
9. Probability of surviving $E = \frac{\text{rate of transition to I}}{\text{rate of transitioning out of E}} = \frac{\kappa}{\mu + \kappa} \Rightarrow \mathscr{R}_0 = \frac{\kappa}{\mu + \kappa} \frac{\beta}{\gamma + \mu}$

While the example above shows the MFMC method with the adjustment applied to a general $SEIR$ model with vital dynamics and a constant population, in Appendix S1 we apply the method (with the adjustment) to a specific example from the literature.

Next let us look at a slightly more complicated model from the literature. Consider the $SEIRD$ model for Ebola dynamics from Weitz and Dushoff (2015). This model is very similar to a traditional $SEIR$, but includes a class of individuals who have died from Ebola and their bodies are still infectious. The model is as follows,

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$$\begin{aligned}
\frac{dE}{dt} &= \frac{\beta_I SI}{N} + \frac{\beta_D SD}{N} - \frac{E}{T_E} \\
\frac{dI}{dt} &= \frac{E}{T_E} - \frac{I}{T_I} \\
\frac{dR}{dt} &= \frac{(1-f)I}{T_I} \\
\frac{dD}{dt} &= \frac{fI}{T_I} - \frac{D}{T_D},
\end{aligned} \tag{4}$$

where $N = S + E + I + R$, D is the class of dead, but infectious individuals, β_I is the transmission rate for contact with infected individuals, β_D is the transmission rate for contact with dead, but infectious individuals, T_E is the average exposed period, T_I is the average infectious period, T_D is the average period of infectiousness after death, and f is the fraction of infectious individuals who die and enter the D class. See Table 5 for a summary of the symbols and parameters used in model 4. Weitz and Dushoff (2015) states that $\mathscr{R}_0 = \beta_I T_I + f\beta_D T_D$. Now we will show how to find $\mathscr{R}_0$ using the MFMC method.

TABLE 5. Symbol and parameter table for model 4

Symbol/parameter	Description
S	Susceptible population
E	Exposed population
I	Infectious and alive population
D	Infectious and dead population
R	Recovered population
N	Total (alive) population ($S + E + I + R$)
β_I	Transmission rate for contact with infectious and alive individuals
β_D	Transmission rate for contact with infectious and dead individuals

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T_D	Average period of infectiousness after death

f	Fraction of infectious and alive individuals that become infectious and dead

Note that this model has two paths to infection: contact with the I class or contact with the D class. These two paths are not in competition and do not have any effect on the other path. Thus, we will formulate two flow systems and compute a basic reproductive number for infections from I, which we will call $\mathscr{R}_I$ and infections from D, which we will call $\mathscr{R}_D$. Then $\mathscr{R}_0$ for model 4 is $\mathscr{R}_I + \mathscr{R}_D$.

First we will calculate $\mathscr{R}_I$:

1. Infection flows through: I, S, E

Infection does not flow through: R

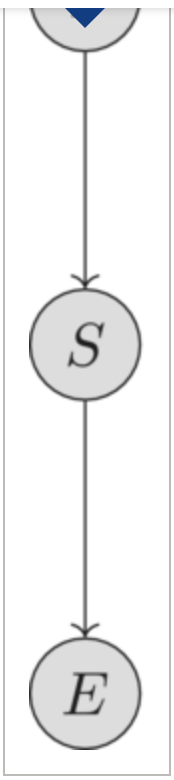
2. $s = I$
3. $t = E$

4.a. $\Phi = \{(I, S)\}$

$\Psi = \{(S, E)\}$

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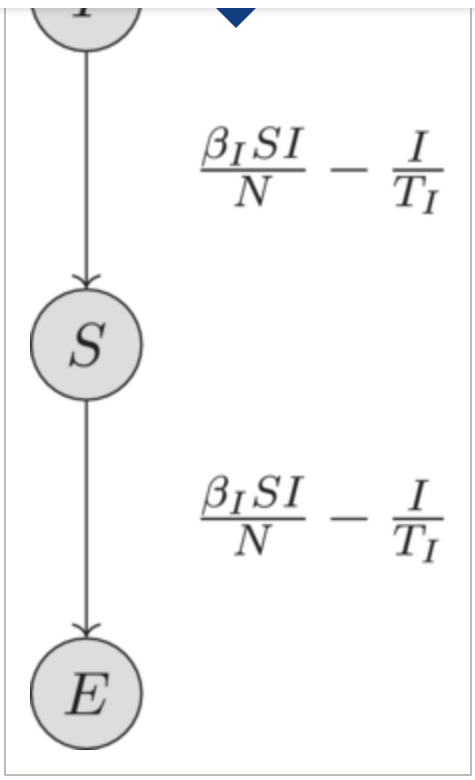


1. 4.b. $w(\Phi) = \frac{\beta_I SI}{N} - \frac{I}{T_I}$

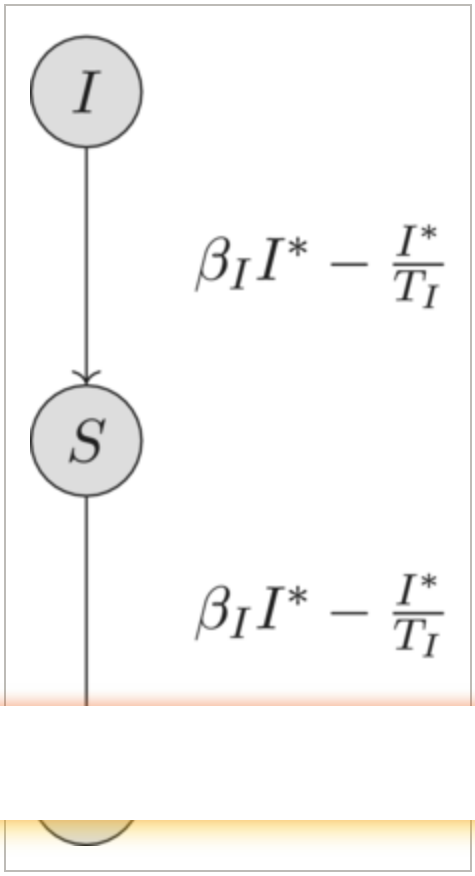
   $w(\Psi) = \frac{\beta_I SI}{N} - \frac{I}{T_I}$

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5. Evaluate near DFE, $(S \approx N, I^* > 0, 0)$:



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7. Minimum cut weight = Maximum flow: $\beta_I I^* - \frac{I^*}{T_I}$

8. Max flow $> 0 \Rightarrow \beta_I I^* - \frac{I^*}{T_I} > 0 \Rightarrow \mathscr{R}_I = \beta_I T_I > 1$

Now we will calculate $\mathscr{R}_D$:

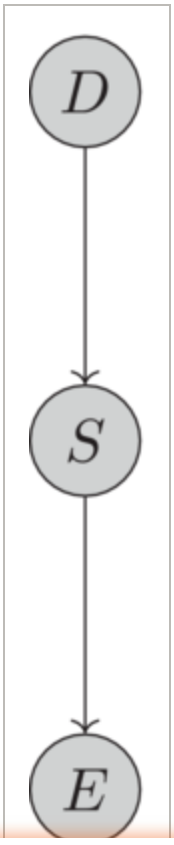
1. Infection flows through: D, S, E

Infection does not flow through: $\emptyset$

2. $s = D$
3. $t = E$

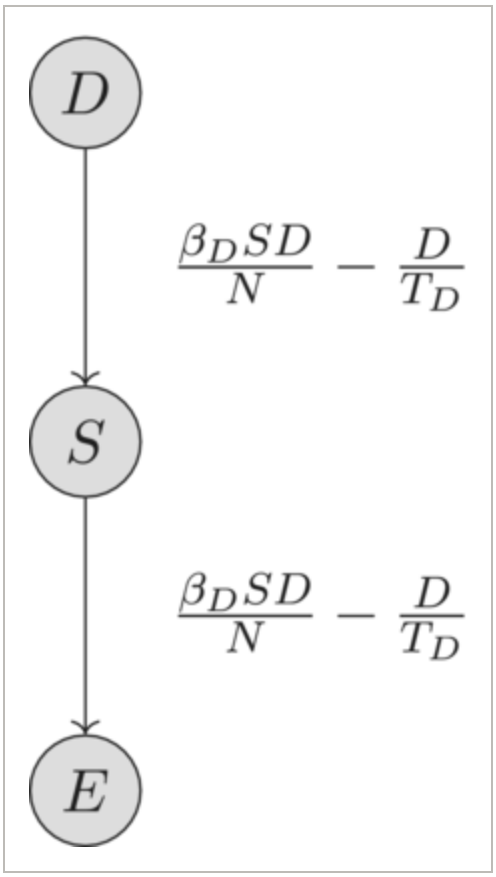
4.a

1. $\Phi = \{(D, S)\}$
   $\Psi = \{(S, E)\}$



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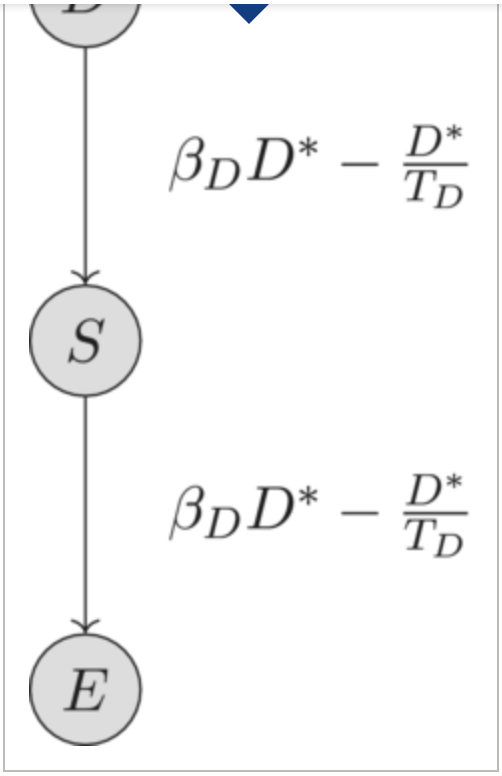
4.b



5. Evaluate near DFE, $(S \approx N, D^* > 0, 0)$:

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6. Cut set: $\{\{(D, S)\}, \{(S, E)\}\}$

7. Minimum cut weight = Maximum flow: $\beta_D D^* - \frac{D^*}{T_D}$

8. Max flow $> 0 \Rightarrow \beta_D D^* - \frac{D^*}{T_D} > 0 \Rightarrow \beta_D T_D > 1$

9. Probability of transitioning to D: $f \Rightarrow \mathscr{R}_D = f\beta_D T_D$

Therefore, $\mathscr{R}_0 = \mathscr{R}_I + \mathscr{R}_D = \beta_I T_I + f\beta_D T_D$. Note that step 9. is applied when calculating $\mathscr{R}_D$. Just as in model 3 where we have to calculate, outside the network, how much disease 'survives' E to go on to I in the next generation, in model 4 we have to calculate how much disease 'survives' the infectious class (transitions to D) to be the source of infection for the $D - S$ infection route. Further this step is needed since the population is not constant, the definition of N does not include D, so we have to consider the amount of 'disease' that leaves the network. We do this by considering the probability an infectious person dies and transitions out of the population into D. By taking that probability into account we capture the amount of infection that comes from disease that left the network.

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models and dynamics. While the initial algorithm presented required a closed system, models 3 and 4 show the algorithm can be adjusted to handle systems that are not strictly closed. Thus,

the NGM method is viable yet complicated, but also allowing us to consider some cases in which the assumptions for the NGM method are not met. Of course, any extensions made to the basic algorithm to increase capability to capture more complicated dynamics may incur additional complexity in calculation, which may then again become prohibitive without sufficient mathematical training. However, even if the actual calculation becomes prohibitive, the use of network formulations as a more naively interpretable representation may still lend valuable insight into the qualitative behaviours of complicated quantitative measures. We believe that one of the most useful aspects of networks are these clear representational formulations and consideration of how other areas of overlap between algebraic systems and algorithmic network formulations may help inspire insights where more traditional formulations alone could not.

6 CONCLUSIONS

Considering a well-studied problem from a different angle can allow for new insights. We have developed a network theoretic approach for calculating and analysing the basic reproductive number and demonstrated how the MFMC method is an equivalent method to the traditional NGM method. By analysing the cut sets of infection flow, we show how the MFMC method can provide valuable, conceptually intuitive information about how to stop an outbreak from progressing. Beyond simplifying access to stable system dynamics, extensions of the methods presented also provide additional tools for analysing transient behaviours. The presented novel MFMC method increases our epidemiological toolkit for teaching and explaining disease dynamics and the basic reproductive number. The availability of this more accessible and intuitive methodology can expand our ability to communicate the perspectives and insights of mathematical modelling to policy makers and help enable real-time response to incipient outbreaks.

AUTHOR CONTRIBUTIONS

Anna Sisk and Nina Fefferman conceived the ideas and designed the algorithm; Anna Sisk and Nina Fefferman showed the equivalence of the methods and analysed the examples; All authors contributed critically to the drafts and gave final approval for publication.

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ideas in this paper, Justin Wright for helpful peer review of the method and manuscript and

CONFLICT OF INTEREST

The authors declare no competing interests.

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DATA AVAILABILITY STATEMENT

Code for the Next Generation Matrix method available from https://doi.org/10.5281/zenodo.6954739 (Sisk & Fefferman, 2022).

Supporting Information

Filename	Description
mee313978-sup-0001-Appendix.pdf PDF document, 977.5 KB	**Appendix S1**

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